

Please attach this cover sheet to your completed homework

Homework 5 (Ch. 11)

ECE 311 – Hang Gao

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Problem	Course outcome	Grade
11.2 (a)	1.c.	/10
11.2 (b)	1.c.	/10
11.5 (a)	1.c.	/10
11.5 (b)	1.c.	/10
11.7 (a)	1.c.	/10
11.7 (b)	1.c.	/10
11.11 (a)	1.c.	/10
11.11 (b)	1.c.	/10
11.12	1.c.	/20
TOTAL:		/100

ABET Assessed Outcomes

1.c. Use appropriate models to formulate solutions related to electric circuit problems.

Homework #5

Due date: Sunday, March 15th by 11:59 PM (Electronically – Canvas)

11.2 A single-phase transformer is rated at 2 kVA, 240/120 V. The transformer is fully loaded by an inductive load of 0.8 power factor lagging. Compute the following:

- Real power delivered to the load
- Load current

11.5 Three single-phase transformers, each is rated at 10 kVA, 400/300 V are connected as wye–delta configuration. Compute the following:

- Rated power of the transformer bank
- Line-to-line voltage ratio of the transformer bank

11.7 Three single-phase transformers are connected as a transformer bank rated at 18 MVA, 13.8 kV(Δ)/120 kV(Y). One side of the transformer bank is connected to a 120 kV transmission line, and the other side is connected to a three-phase load of 12 MVA at 0.8 lagging power factor. Compute the following:

- Turns ratio of the transformer bank
- Line current at the 120 kV side

11.11 The transformer shown in Figure 11.6 consists of one primary winding and two secondary windings. The numbers of turns of the windings are: $N_1 = 10,000$; $N_2 = 5,000$, and $N_3 = 1,000$. A voltage source of 120 V is applied to the primary winding. The load of winding N_2 consumes 600 W and 300 VAR inductive power. The load of winding N_3 consumes 24 W and 36 VAR capacitive power. Compute the following:

- Voltages of the secondary windings
- Currents of all windings

11.12 A single-phase transformer has three windings. The primary winding N_1 carries a current of $I_1 = 10$ A. One of the secondary windings has 3000 turns and carries 2 A, and the other secondary winding has 6000 turns and carries 1 A. All currents are in phase. Compute the number of turns in the primary winding.

- 11.2 The reference information allows us to assume that at full load (at maximum capacity), the given transformer will deliver $S = 2^{\text{kVA}}$ of apparent power, with a voltage ratio of $\frac{E_1}{E_2} = \frac{240^{\text{V}}}{120^{\text{V}}} = 2$. Assuming all values are given in RMS form,

- a) Using the formula for power factor,

$$\text{P.F.} = \frac{P}{S},$$

we may calculate real power P as

$$P = 0.8 \times 2^{\text{kVA}} = 1.6^{\text{kW}}.$$

- b) Using the formula for RMS current,

$$I = \frac{S}{E},$$

we may calculate load current as

$$I_{\text{load}} = \frac{2^{\text{kVA}}}{120^{\text{V}}} = 16.67^{\text{A}}.$$

- 11.5 a) For three transformers rated at $S = 10^{\text{kVA}}$, we may find the total rated power of the transformer bank as

$$S_{\text{total}} = 10^{\text{kVA}} \times 3 = 30^{\text{kVA}}.$$

- b) For a wye configuration, the line-to-line voltage is $\sqrt{3}$ times greater than the line-to-neutral voltage, so the voltage across N_1 is

$$V_{ab} = 400^{\text{V}} \times \sqrt{3} = 692.82^{\text{V}}.$$

In the delta configuration, on the other hand, the voltage across N_2 is the same as the line-to-line voltage. Therefore, the line-to-line voltage ratio is

$$\frac{N_1}{N_2} = \frac{692.82^{\text{V}}}{300^{\text{V}}} = 2.31$$

- 11.7 a) To find the turns ratio, we may simply use the same assumptions stated in the previous question, such that the turns ratio is

$$\frac{N_1}{N_2} = \frac{(120/\sqrt{3})^{\text{kV}}}{13.8^{\text{kV}}} = 5.02,$$

assuming that this is a *step-down* transformer bank.

- b) Finding line current as a function of power on the load side,

$$I = \frac{12^{\text{MVA}}}{(120 \times \sqrt{3})^{\text{kV}}} = 57.74^{\text{A}}.$$

- 11.11 a) First, we may compute the voltage per turn on the primary side as

$$V_T = \frac{E_1}{N_1} = \frac{120^{\text{V}}}{10,000} = 12^{\text{mV}},$$

and then compute the voltage on the two secondary windings as

$$E_2 = V_T \times N_2 = 12^{\text{mV}} \times 5,000 = 60^{\text{V}}$$

$$E_3 = V_T \times N_3 = 12^{\text{mV}} \times 1,000 = 12^{\text{V}}$$

- b) Computing the current in each winding as a function of the apparent power S and the winding voltage,

$$I_1 = \frac{678^{\text{VA}}}{120^{\text{V}}} = 5.64^{\text{A}}$$
$$I_2 = \frac{671^{\text{VA}}}{60^{\text{V}}} = 11.18^{\text{A}}$$
$$I_3 = \frac{43^{\text{VA}}}{12^{\text{V}}} = 3.61^{\text{A}}$$

11.12 Using the equivalence

$$N_1 I_1 = N_2 I_2 + N_3 I_3,$$

$$N_2 I_2 = 3000 \times 2^{\text{A}} = 6000$$

$$N_3 I_3 = 6000 \times 1^{\text{A}} = 6000,$$

so that $N_1 I_1 = 12,000$, and

$$N_1 = \frac{12,000}{I_1} = 1,200.$$